

Automatic Emotion Detection from Facial Features Extraction and Machine Learning Models

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Abstract- In the domain of Interactive Systems (IS), machine learning (ML) emerges as a profound, user-friendly, and accurate approach to enhancing emotional intelligence in digital systems. This research work focuses on facial emotion detection, an important but often-neglected aspect of affective computing. Models used for the assessment include Multinomial Naïve Bayes (MNB), Decision Trees (DT), K-Nearest Neighbours (KNN), Random Forest (RF), Support Vector Machines (SVMs), XG Boost (XGB), and Logistic Regression (LR). By employing the FER2013 dataset and MediaPipe's (MP's) facial recognition technology, this research extracts 478 3D facial landmarks to test. The best Acc in emotion detection is found using LR with an accuracy of 65.57% followed by XGB with an accuracy of 61.24%. This study thus will help as a practical application toward making improvements in IS in matters regarding health, marketing, or even security.

Keywords- Emotion Recognition, Facial Images, Dataset, ML, Deep Learning (DL), Supervised Learning (SL)

I. INTRODUCTION

Facial emotion detection is an essential component of improving IS and plays a role in various fields like healthcare, marketing or security. Using MP's real-time facial recognition we get the key point coordinates, this was used to classify (6 emotions - happy, sad, surprise, angry, disgust and neutral) of emotion excited. MP is fast and accurate in extracting facial features, which makes it a trendy choice. Ekman et al. [1], a noted psychologist, categorises facial expressions into six fundamental emotions. Since then, numerous researchers have uniformly acknowledged these basic emotions in FER (Facial Emotion Recognition) study. The six fundamental emotions are surprise, grief, happiness, disgust, anger, and neutral.

FRS became a necessity for automatic identification, thus fueling research in biometrics and digital security. Pattern recognition and artificial intelligence fueled the technological development of IS technologies [2]. A significant subset in this domain is Automatic Emotions Detection from Facial Features Extraction; this becomes a significant area as it makes the machines interpret and react to human sentiments. This paper surveys and reviews the critical elements of FER,

including facial feature extraction, emotional databases, and classifier algorithms, as an example of how FER progresses in real-time, emotion-aware systems toward more personalized and intuitive interaction [3]. This research paper aims at developing and implementing an efficient facial emotion detector based on advanced facial recognition technology of MP, while classifying six basic emotions: happy, sad, surprise, angry, disgust, and neutral.

II. LITERATURE SURVEY

This is essentially a multidisciplinary field of research that combines AI, ML, and signal processing in order to classify emotions based on inputs such as facial expressions, speech, EEG signals, text, and physiological markers. Applications of human emotion detection have been labeled as pertinent in healthcare, IS, and customer services. In extensive study, speech-based emotion detection has been looked into highly, with Zhang et al. [4] proposing a multi-view learning framework for the sake of A_{cc} . Park et al. [5] reported challenges on speech detection in real-world environments, noisy conditions. The algorithms used to carry out research into this domain are not yet robust enough to improve the underlying mechanism of emotion detection. Advances in DL have played a great role in influencing EEG-based emotion detection. Al-Nafijan et al. [6] used neural networks showing good A_{cc} in classification. Nonetheless, processing EEG data is still tough due to noise and limited datasets.

Also, Calix et al. [7] increased text-based emotion detection by applying SL techniques, but one of the problems that are identified and have been reported concerning Hossain et al. [8] is about minimal context as well as detection difficulties especially regarding sarcasm. The multimodal emotion detection including speech, facial expressions, and physiological signals can be considered high promise because of its capability to integrate the complementary data sources. The model from Jaques et al. [9] has improved performance through the learning of latent factors from multi-type data. Wang et al. [10] has expanded the above using hybrid techniques enhancing generalization and giving high-generation capacities, which make it suitable for broader

applicability in real-world environments. Body language recognition, as reported in work by Berthouze et al. [11], and facial expression analysis via AffectNet from Mollahosseini et al. [12], are contemporary studies. Apart from the application of biophysical signals such as heart rate and skin conductance as emotion detection means when facial or speech data is unavailable, Song et al. [13] mention an important area.

Kirar et al. [16] have provided an excellent systematic review throughout a plethora of sources and provided a future roadmap for work. Generally, multimodal approaches are best suited for accurate emotion detection; however, future research faces several significant obstacles, such as the noise levels in real-world data and advancements in text-based methods. Table I, which is provided below, shows the highest level of A_{cc} that each paper's methodology managed to achieve. It also gives a thorough summary of all the literature that has been reviewed in prior publications.

TABLE I. SUMMARY OF THE LITERATURE.

Refer.	Year	Methods/Models	A_{cc}
[4]	2020	Multi-view learning (SVM, GMM)	54.3%
[5]	2017	CNN, LSTM	58%
[6]	2013	Naive Bayes, SVM, Decision Trees	52%
[9]	2016	Neural networks, latent factor modelling	59.4%
[10]	2018	RF, k-NN	57.8%
[12]	2019	CNN	59.6%
[14]	2021	CNN	63%
[15]	2022	Lightweight CNN	61%
[6]	2016	SVM, deep neural networks	56.2%
[7]	2018	SVM, CNN	55%
[9]	2019	CNN (visual), RNN (auditory)	58.5%
[10]	2019	SVM, Naive Bayes, LSTM	57%

III. PROPOSED METHODOLOGY

A. Dataset

The FER2013 dataset [17] is one of the primary sources for FER. It contains 35,887 grayscale images with a pixel resolution of 48x48 that are categorized into seven emotions: angry, disgust, fear, happy, sad, surprise, and neutral. It addressed some of the standard concerns in the FER literature, namely, privacy and small sample sizes. The FER2013 dataset splits into training (28,709 images), public test (3,589 images), and private test (3,589 images), making it possible to train models in an extremely robust way and check them up. Providing an extensive set of such labeled data, FER2013 is helpful to ML practitioners in training and testing models of accurate emotion recognition. Its availability encourages the creative collaboration within the research community that enhances further progress within computer vision and affective computing. Fig 1 represents the sample images of the FER2013 dataset.



Fig. 1. Typical sample images: (a) Happy, (b) Surprised, (c) Sad, (d) Disgust, (e) Angry and (f) Neutral.

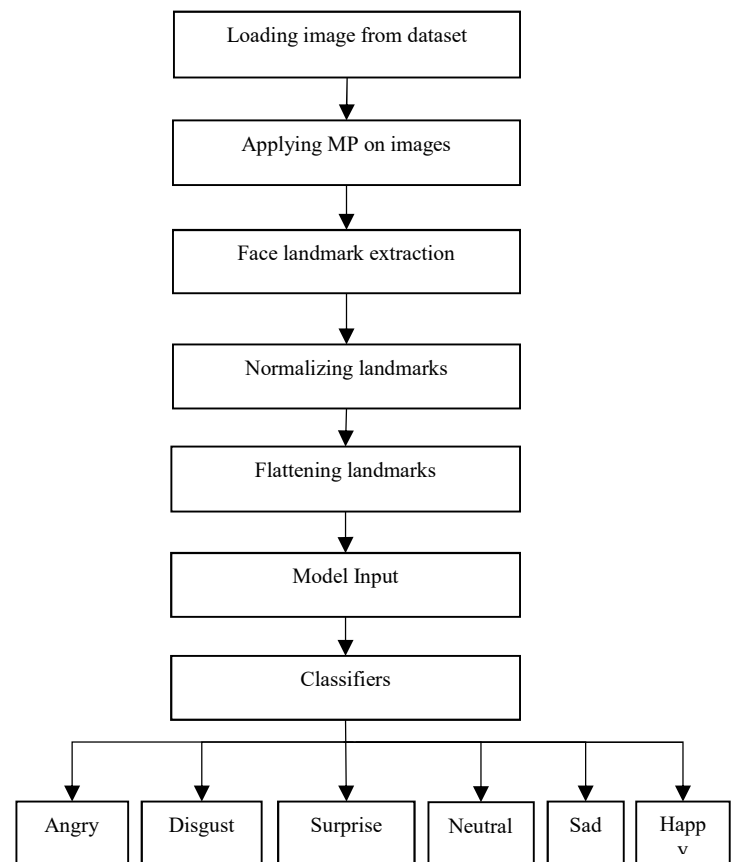


Fig. 2. Representation of proposed method.

B. Methodology

The techniques used in this experiment are presented in Figure 2. To differentiate facial emotions, we suggest the usage of pre-trained ML models combined with feature extraction methods. Before that, to normalize the input to the model, image processing and the detection of facial landmarks takes place. The extracted features are passed through the seven ML models [18-22], which are used in the process of detecting emotions:

- i. MNB
- ii. Decision Trees
- iii. KNN
- iv. RF
- v. SVMs
- vi. XGB
- vii. LR

C. Feature Extraction and Normalization

For this experiment, we utilize MP's face mesh model to extract 478 3D facial landmarks from each input image. The x, y, and z coordinates returned by the face mesh model are then concatenated into a feature vector. For each image, we have a resulting feature vector of dimensionality 1434, corresponding to 478 landmarks $\times$ 3 coordinates.

Mathematically, the feature vector for the i -th image, denoted as F_i , is represented in the following manner:

$$F_i = [(x_1, y_1, z_1), (x_2, y_2, z_2), \dots, (x_{478}, y_{478}, z_{478})] \quad (1)$$

where (x_j, y_j, z_j) represents the 3D coordinates of the j -th landmark. This high-dimensional feature vector serves as the input for further preprocessing and classification tasks.

Then, min-max normalization is used to scale the features so that all of them are of the same range. Min-max normalization differs from z-score standardization as it tries to standardize the data so that all values fall in the range between 0 and 1, while z-score standardization sets the data so that the mean is 0 and the normal deviation is 1. In min-max normalization, each attribute is adjusted to some value ranging from 0 to 1, which is very crucial because many ML algorithms are sensitive to the scales of the features utilized, particularly distance-based methods such as KNN. The mathematical equation for the min-max normalization of the attribute is:

$$z = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (2)$$

Where: Normalized value of the feature is indicated by z , Original value of the feature is indicated by x_i , Min. value of the feature across all samples is $x_{\min}$, Max. value of the feature across all samples is $x_{\max}$.

In every feature f of the feature vector, the normalization is performed as follows:

$$\begin{aligned} x_{\min_f} &= \min(f_i), \text{ where } \min(f_i) \text{ depicts the } \min^m \text{ value of feature } f \text{ across the samples,} \\ x_{\max_f} &= \max(f_i), \text{ where } \max(f_i) \text{ depicts the } \max^m \text{ value of feature } f \text{ across the samples,} \\ z_f &= \frac{f - x_{\min_f}}{x_{\max_f} - x_{\min_f}}, \text{ } z_f \text{ depicts the normalized value of feature } f. \end{aligned}$$

Normalizing the feature values to a range [0, 1] neutralizes the varying magnitudes between the features. Consequently, this will improve the performance of the learning algorithms used in the emotion classification. That's because no particular feature will dominate due to scale differences. That's crucial since the facial landmark coordinates are diverse.

D. Landmark Flatenning

Landmark flattening refers to the conversion of 3D coordinates from facial landmarks to a one-dimensional vector. It expresses (x, y, z) in a one-dimensional form. Flattening is critical for performing tasks involving emotion detection by attaining efficient training and classification.

E. Training and classification

In this study, classification models utilized for emotion detection from facial landmarks were MNB, Decision Trees, KNN, RF, SVM and XGB. The model was fit and trained on the data source with an 80-20 train-test split while trying a different hyperparameter and iteration to get the best performance. Naïve Bayes needs smoothing factors to be tuned for each of them, Decision Trees with maximum depths and criteria need to be adjusted, KNN used different k-values, RF requires the number of trees to be set for specification, SVM includes choosing of kernel functions to be applied, whereas XGB requires learning rates and boosting rounds to be determined. Through systematic tuning and evaluation of the hyperparameters, we have effectively and robustly classified emotions based on facial landmarks while optimizing the A_{cc} of our model and efficiency in its computation.

F. Performance Matrices

Equations (3-6) have been used in this work to evaluate the Accuracy (A_{cc}), Precision (P_{cs}), Recall (R_c) and $f1_{score}$ performance matrices [18-22]:

$$A_{cc} = \frac{T_N + T_P}{F_N + F_P + T_N + T_P} \quad (3)$$

$$P_{cs} = \frac{T_P}{F_P + T_P} \quad (4)$$

$$R_c = \frac{T_P}{F_N + T_P} \quad (5)$$

$$f1_{score} = \frac{2 \times P_{cs} \times R_c}{P_{cs} + R_c} \quad (6)$$

Where:

T_P : True Positive, no. of instances predicted correctly as +ve.
 T_N : True Negative, no. of instance predicted correctly as -ve.
 F_N : False Negative, the no. of instances incorrectly predicted -ve.
 F_P : False Positive, the no. of instances incorrectly predicted +ve.
 Assessing how well classifications operate is crucial. It's done through A_{cc} , P_{cs} , and R_c metrics for understanding how well our emotion detection model works. P_{cs} is the correctness of positive predictions, while R_c is defined by how correctly the model identifies true positives. Overall, the choice of the right model is ensured to further improve performance in accurately classifying emotions, especially where the datasets are imbalanced. if datasets are imbalanced; for that matter, consistency across various scenarios will not yet be achieved.

TABLE II. COMPARATIVE PERFORMANCE OF PROPOSED CLASSIFICATION MODELS FOR FER

S. No.	Classifiers	A_{cc}	Est. P_{cs}	Est. R_c	Est. f1 Score
1	MNB	38.73%	0.40	0.39	0.39
2	Decision Trees	45.48%	0.46	0.45	0.45
3	KNN	46.26%	0.47	0.46	0.46
4	RF	57.78%	0.58	0.58	0.58
5	SVM	59.32%	0.60	0.59	0.59
6	XGB	61.24%	0.62	0.61	0.61
7	LR	65.57%	0.65	0.66	0.65

IV. RESULTS AND DISCUSSION

In this paper, for all classifiers, A_{cc} , P_{cs} , R_c , and f1 are used as performance metrics. Results of experiments are reported in Table II: Highest A_{cc} obtained from all the experiments is achieved by LR with a success rate of 65.57%. This is followed by XGB at 61.24% and SVM at 59.32%. Other models, such as RF (57.78%), DT (45.48%), KNN (46.26%), and MNB (38.73%), have different states of effectiveness and reflect the pros and cons of each approach related to emotion classification. Lower A_{cc} of MNB is because of its inherent assumption of independence between features which usually does not satisfy in complex data with dependent features like facial landmarks. Decision Trees and KNN were weak as well because those algorithms are prone to overfitting and therefore don't perform so well with such high-dimension data. RF, being stronger than Decision Trees but weaker than the other two, made a few points as it was because of data and feature representation and that is why the A_{cc} was not as good as XGB and LR did.

XGB and LR happen to especially excel in emotion detection due to their unique strengths. XGB utilizes the concept of ensemble learning, employing a combination of more than one weak learner - decision trees that come together to form a stronger model, thereby reducing bias and variance. Its inbuilt techniques for regularization through L1 and L2 lessen overfitting, bettering generalization on unseen data.

The output is sturdier in actuality due to its functionality in handling missing values. On the other hand, LR has a simple and intuitive model by describing linear dependency between features and classes. It can easily provide probability of class membership useful for decision making in multi-class cases. Finally, it's also friendly to compute when data size is small. Putting all these advantages together, XGB and LR outperform the rest in emotion detection tasks.

The output of the LR model in Table III shows that the maximum number of iterations may be changed to impact A_{cc} in different ways. From the above, it can be seen that increasing the maximum number of iterations from 250 to 1000 showed increments in the A_{cc} of the model at a linear and steady pace until it reached its best point at 65.57%. However, increments greater than this and to 1100 and 1250 iterations showed a slight decline in performance, indicating that with too many iterations, the same overtraining can set in. The different maximum iterations settings did not change the performance of the model drastically, which shows how effective convergence has been at these settings. This trend brings on the importance of parameter tuning to further optimize model performance since the selection of iterations can vary and thereby affect convergence and A_{cc} . Once again, in the case of Emotion detection, the selected iteration has a great influence on the convergence and A_{cc} of the LR algorithm.

TABLE III. VARIATION IN A_{cc} WITH HYPERPARAMETER TUNING

C value	Penalty	Solver	Class weight	Max Iterations	A_{cc}
1	L2	lbfgs	none	250	63.18%
1	L2	lbfgs	none	500	64.76%
1	L2	lbfgs	none	750	65.11%
1	L2	lbfgs	none	1000	65.57%
1	L2	lbfgs	none	1100	65.07%
1	L2	lbfgs	none	1250	63.35%
1	L2	lbfgs	none	1500	64.34%

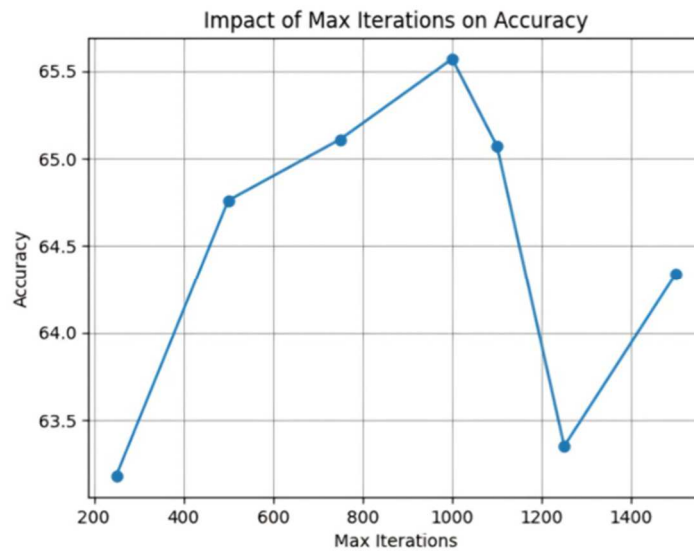


Fig. 3. Impact of iteration on Acc for LR .

The graph (fig 3) depicts the interaction between maximum iterations and model A_{cc} for LR. From 200 to 1000 iterations, it is observed that with the rise in the number of iterations, the capacity of the model being more accurate increases dramatically to reach a maximum of 65.57% with 1000 iterations. Therefore, in this scenario, it may be inferred that more iterations allow the algorithm to be better converged such that model parameters get properly optimized. However, beyond this point, a significant drop in A_{cc} is shown-especially at 1250 and 1400 iterations, meaning possible overfitting or diminishing returns in A_{cc} gains. In any case, this poses an important issue with respect to the right choice of iterations for optimum model complexity with regard to performance in tasks like emotion detection.

The result depicted in Table IV for the emotion classification models are summarized in terms of P_{cs} , R_c , and f1 score, and support for each class. The "Angry" class has a $P_{cs} = 0.55$ and $R_c = 0.49$, resulting in an f1 score = 0.51. The "Disgusted" class display a lower $P_{cs} = 0.56$ and a significantly low $R_c = 0.12$, leading to an f1 score = 0.20. In contrast, model excels in the "Happy" class, with $P_{cs} = 0.79$, a $R_c = 0.85$, and an f1 score = 0.82, demonstrating exceptional results. The "Neutral" and "Sad" classes exhibit P_{cs} and R_c values nearly

equal to 0.56 and 0.54, respectively, resulting in similar f1 scores of 0.60 and 0.51. The model attains a comprehensive result with $A_{cc} = 0.66$, with macro averages = 0.63 for P_{cs} , 0.56 for R_c , and 0.57 for f1 score. The proposed method (PM) is better in comparison to existing as shown in Fig. 4.

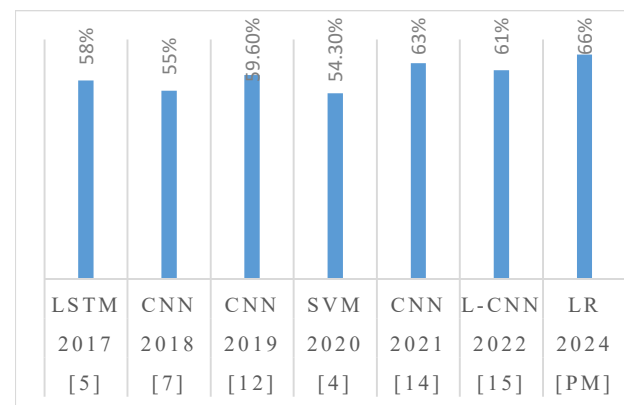


Fig. 4. Comparison of methods.

TABLE IV. CLASSIFICATION REPORT FOR LR (MAX. ITERATION = 1000)

Class	P_{cs}	R_c	f1 score	support
Angry	0.55	0.49	0.51	708
Disgusted	0.56	0.12	0.2	81
Happy	0.79	0.85	0.82	1363
Neutral	0.56	0.63	0.6	992
Sad	0.54	0.48	0.51	886
Surprised	0.78	0.77	0.78	576
A_{cc}	0.66	0.66	0.66	4609
Avg macro	0.63	0.56	0.57	4609
Avg weighted	0.65	0.66	0.65	4609

V. CONCLUSION

Emotion detection is now a very essential aspect to understand human behaviour and is applicable to many domains. Here, in the paper, we have proposed a holistic approach regarding emotion detection with the help of MP toward feature extraction from facial images. For achieving high A_{cc} toward identification of different kinds of emotional states, we used a variety of ML models. The obtained results suggested the best-performing model demonstrated 66% A_{cc} in the case; this obviously proves the validity of our approach. In particular, various models working well on different emotions and its subcategories indicated that the proposed approach could effectively recognize different emotions and outperform state-of-the-art approaches in certain cases. This work will help contribute to both the area of IS as well as the field of mental health monitoring systems by integrating advanced image processing techniques with ML for emotion recognition.

REFERENCES

- [1] P. Ekman and H. Oster, "Facial expressions of emotion," *Annual Review of Psychology*, vol. 30, no. 1, pp. 527–554, 1979.
- [2] K. F. Azizan Illiana, "FER: A Brief Review," in *International Conference on Sustainable Engineering, Technology and Management 2018 (ICSETM-2018)*, 2020.
- [3] R. Shyam, "Automatic Face Recognition in Digital World," *Advances in Computer Science and Information Technology (ACSIT)*, vol. 2, no. 1, pp. 64-70, 2015.
- [4] L. Zhang, J. Pan, and Y. Wu, "A Multi-View Learning Framework for Emotion Detection from Speech," *IEEE Access*, vol. 8, pp. 95648-95657, May 2020.
- [5] T.J. Park and C.M. Lin, "Low Accuracy of Emotion Detection Using Voice Features in Real-World Environments," *Speech Communication*, vol. 99, pp. 34-45, Oct. 2018.
- [6] A. Al-Nafjan, M. Hosny, and R. Al-Ohali, "EEG-based Emotion Detection Using DL Techniques," *Frontiers in Neuroscience*, vol. 11, pp. 562-576, June 2017.
- [7] R. Calix, C. Montalvo, and P. Gupta, "Detecting Emotion in Text Messages Using SL Techniques," *Journal of Artificial Intelligence and Soft Computing Research*, vol. 3, no. 4, pp. 301-310, Oct. 2013.
- [8] A. Hossain and N. M. Sakib, "Challenges in Emotion Detection Using Text-based Approaches," *IEEE Intelligent Systems*, vol. 34, no. 2, pp. 20-27, Apr. 2019.
- [9] N. Jaques, M. Taylor, and T. Sano, "Predicting Affect from Multimodal Data Using Learned Latent Factors," *ACM Transactions on Interactive Intelligent Systems*, vol. 5, no. 4, pp. 25-36, Nov. 2016.
- [10] Y. Wang, M. Hu, and J. Li, "Multimodal Emotion Detection Using Hybrid Techniques," *Pattern Recognition Letters*, vol. 124, pp. 123-130, Mar. 2019.
- [11] S. Berthouze and J. Hayashi, "Emotion Recognition from Bodily Expressions for Adaptive Systems," *IEEE Transactions on Affective Computing*, vol. 9, no. 2, pp. 185-200, Apr. 2018.
- [12] D. Mollahosseini, B. Hassani, and M. H. Mahoor, "AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild," *IEEE Transactions on Affective Computing*, vol. 10, no. 1, pp. 18-31, Jan. 2019.
- [13] H. Song, D. Lee, and K. Choi, "Emotion Detection Using Biophysical Signals," *Sensors*, vol. 16, no. 8, pp. 1254-1269, Aug. 2016.
- [14] Y. Wang, F. Gao, and L. Chen, "FER Based on Low-Resolution Images Using CNN," (2021)
- [15] J. Liu, M. Li, and K. Wang, "Robust FER Under Partial Occlusions with Lightweight CNNs," (2022)
- [16] B.S. Kirar, J. Madavi, A. Prajapati, L. Solanki, and P. Newalkar, Advancements in emotion recognition: systematic review and research roadmap, In: Rawat, S., Kumar, A., Raman, A., Kumar, S., Pathak, P. (eds) *Proceedings of Third International Conference on Computational Electronics for Wireless Communications. ICCWC 2023. Lecture Notes in Networks and Systems*, vol 962, 2025. Springer, Singapore. https://doi.org/10.1007/978-981-97-1946-4_11.
- [17] Y. Khairuddin and Z. Chen, "FER: State of the Art Performance on FER2013," *arXiv preprint arXiv:2105.03588*, May 2021, doi: 10.48550/arXiv.2105.03588.
- [18] B.S Kirar, G.R.S. Reddy, and D.K. Agrawal, Glaucoma detection using SS-QB-VMD based fine sub band images from fundus images, *IETE Journal of Research*, Vol. 69, Issue 8, pp. 4909-4920, 2023.
- [19] B.S. Kirar, D.K. Agrawal, and S. Kirar, Glaucoma detection using image channels and discrete wavelet transform, *IETE Journal of Research*, Vol. 68, Issue 6, pp. 4421-4428, 2022.
- [20] D.K. Agrawal, B.S. Kirar, and R.B. Pachori, Automated glaucoma detection using quasi-bivariate variational mode decomposition from fundus images, *IET Image Processing*, Vol. 13, Issue 13, pp. 2401-2408, November 2019.
- [21] B.S. Kirar and D.K. Agrawal, Computer aided diagnosis of glaucoma using discrete and empirical wavelet transform from fundus images, *IET Image Processing*, Vol. 13, Issue 1, pp. 73-82, January 2019.
- [22] S. Dixit, B.S. Kirar, and D.K. Agrawal, Performance analysis of DL models for accurate glaucoma detection, *8th IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS)*, 24-25 February 2024, Bhopal, India, pp. 1-6, April 2024.